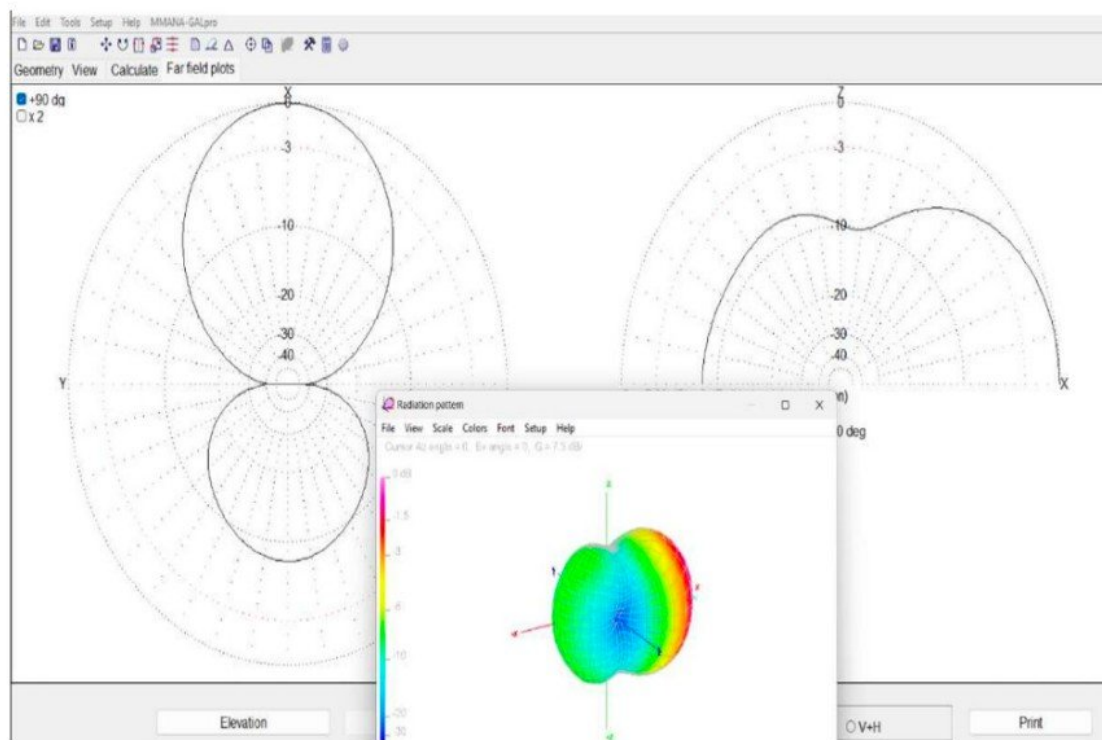
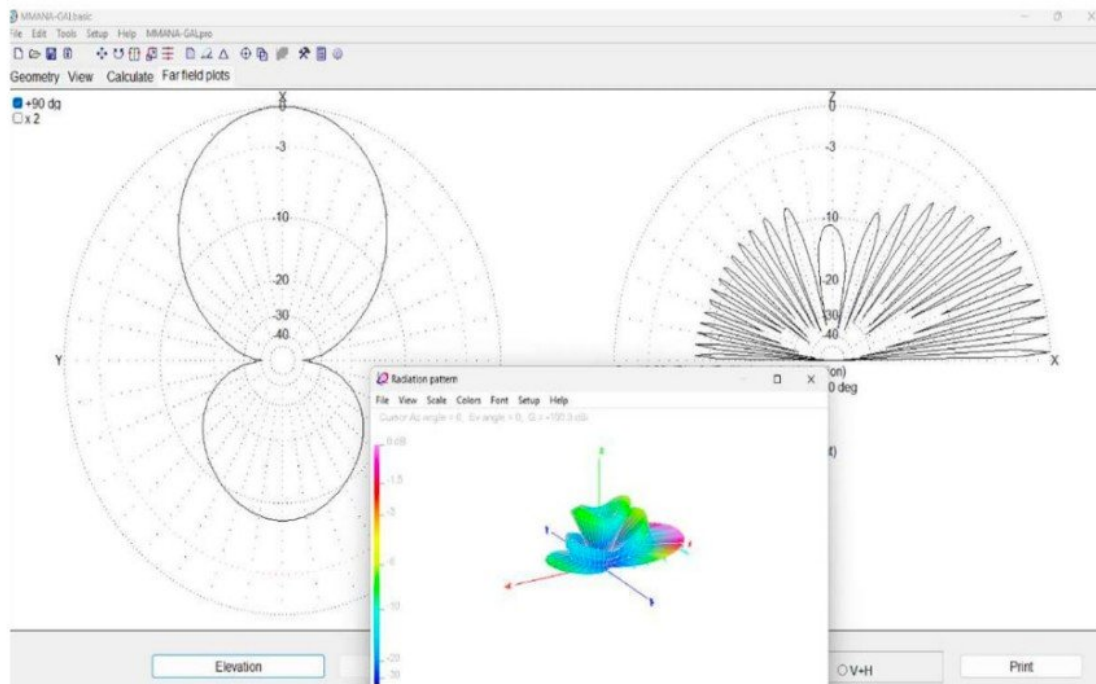


5a. Design and Optimization of Wire Antenna using MMANA-GAL Software





5b. Design and Optimization of Wire Antenna using MMANA-GAL Software

MMANA-GALpro

File Edit Tools Setup Help MMANA-GALpro

Geometry View Calculate Far field plots

Yagi 5 Element

Freq 143.2 MHz

Ground

☒ Free space

☐ Perfect

☐ Real

Add height 15 m

Material Al pipe

WAVE LENGTH = 2.094 (m)
TOTAL PULSE = 103
FILL MATRIX
FACTOR MATRIX
PULSE U (V) I (mA) Z (Ohm) SWR PWR(WT)
w2c 10.00+0.000 103.5+06.51 44.59+21.02 1.58 1.8348
POWER = 1.83 WT
CURRENT DATA
FAR FIELD (Pin = 1.8348 WT)
NO FATAL ERROR(S)
0.98 sec

No.	F (MHz)	R (Ohm)	μX (Ohm)	SWR 50	Gh dBd	Ga dBi	F/B dB	Elev.	Ground	Add H.	Polar.
22	143.2	44.59	-21.02	1.58	5.11	7.26	7.66	---	Free	---	hori.
21	143.2	44.59	-21.02	1.58	5.11	7.26	7.66	---	Free	---	hori.
20	143.2	44.59	-21.02	1.58	5.11	7.26	7.66	---	Free	---	hori.
19	143.2	44.59	-21.02	1.58	5.11	7.26	7.66	---	Free	---	hori.
18	143.2	44.59	-21.02	1.58	5.11	7.26	7.66	---	Free	---	hori.
17	143.2	44.59	-21.02	1.58	5.11	7.26	7.66	---	Free	---	hori.
16	143.2	44.44	-20.76	1.57	---	13.28	7.82	2.0	Perfect	15.0	hori.
15	143.2	44.44	-20.76	1.57	---	13.28	7.82	2.0	Perfect	15.0	hori.
14	143.2	44.44	-20.76	1.57	---	13.28	7.82	2.0	Perfect	15.0	hori.

Start Optimization Optimization log Plots Wire edit Element edit

